Week 10

1. Summary

This week the model stays fixed, the tuned logistic regression selected in Week 9 (C = 3), and the data is improved around it. Error analysis on the training data showed that at the 0.5 cutoff the model misses 86 of 187 responders, mostly customers with no previous campaign history, and that a few features contain extreme values. We tested three kinds of data improvement: capping extreme values, oversampling responders, and adding features aimed at the error groups. Only capping the continuous features improved cross-validated PR-AUC (+0.0044), so it was selected.

On the updated validation dataset the Week 9 model and the Week 10 model are statistically tied (PR-AUC 0.730 and 0.727). We selected the Week 10 model, trained on the capped data, because it is better in cross-validation and on four other validation metrics, and it is less sensitive to extreme values. On the test set the final model reaches a PR-AUC of 0.678 and captures 85.7% of responders in the top 40% of customers. The main limit on performance is missing signal for customers with no campaign history, not data quality.

2. Datasets

The 1,608 original training customers were split once into a training dataset (80%) and a validation dataset (20%), stratified by response so the response rate is the same in both. Weeks 6 to 9 used cross-validation inside the training data; the split lets every dataset be reported separately this week. The 412-customer test dataset was not used for any decision.

Table 1. Datasets

Dataset	Customers	Responders	Response rate
Training	1,286	187	14.5%
Validation	322	47	14.6%
Test	412	63	15.3%

3. Error Analysis and the Data Improvements It Led To

Predictions for every training customer were made by models that never saw that customer (5-fold cross-validation), so the errors are not hidden by memorization. All features are standardized, so group means below are in standard deviation units.

What the errors look like.

- **Many responders are missed at the 0.5 cutoff.** The model finds 101 of 187 responders (54.0%), misses 86 and wrongly flags 41 non-responders, a precision of 71% (Table 2).
- **The probabilities themselves are reliable.** Predicted and actual response rates agree across the ten probability deciles; the top decile is predicted at 0.80 and 74% responded, and the ninth decile is predicted at 0.36 and 42% responded. The low recall comes from the cutoff and from missing signal, not from miscalibration.
- **Missed responders look like non-responders.** Compared with the responders the model finds, they have far fewer previous campaign acceptances, lower income per person and more deal purchases (Table 3).
- **False alarms look like responders.** Compared with correct non-responders, they have more previous acceptances, more recent purchases, higher spending and higher income per person (Table 4).
- **Errors are concentrated in low-income and low-spend customers.** In the lowest income-per-person quartile the model finds only 27% of responders, against 71% in the highest quartile (Table 5).
- **The no-history group drives the misses.** 1,024 of the 1,286 training customers (80%) have never accepted a previous campaign and respond at only 7.2%. That group contains about 74 of the 187 responders, roughly 40% of all of them, and on the data we have they are hard to tell apart from non-responders.

Table 2. Out-of-fold confusion matrix on the training data (0.5 cutoff)

	Predicted: no response	Predicted: response
Actual: no response	1,058	41
Actual: response	86	101

Table 3. Missed responders (FN) vs found responders (TP): five largest differences

Feature	Mean, missed (FN)	Mean, found (TP)	Std. mean difference
Total_Prev_Accepted	0.293	1.720	-1.007
Income_per_Person	0.163	1.147	-0.770
NumDealsPurchases	0.240	-0.471	0.669
Lives_Alone	0.151	0.436	-0.654
MntMeatProducts	0.236	0.738	-0.561

Table 4. False alarms (FP) vs correct non-responders (TN): five largest differences

Feature	Mean, false alarm (FP)	Mean, correct (TN)	Std. mean difference
Total_Prev_Accepted	0.784	-0.243	0.891
Recency	-0.690	0.107	-0.868
Income_per_Person	0.655	-0.182	0.757
Total_Spend	0.511	-0.133	0.700
MntMeatProducts	0.527	-0.132	0.697

Table 5. Error rates by income-per-person quartile (training data, out-of-fold)

Income per person (standardized)	Customers	Response rate	Recall (responders found)	Mean predicted probability
Lowest quartile (-1.225 to -0.670)	322	0.081	0.269	0.086
Second quartile (-0.670 to -0.326)	321	0.093	0.367	0.089
Third quartile (-0.326 to 0.349)	321	0.103	0.394	0.108
Highest quartile (0.349 to 3.825)	322	0.304	0.714	0.305

Data audit. The training data has one duplicated customer, no contradictory duplicates, no missing values and no constant features. Skew is mild (the largest is 1.79, for income per person), and because the features are standardized and contain negative values, a log transform does not apply. A few features contain extreme values: income reaches 5.12 although its 99th percentile is 1.99, web visits per month reaches 6.24 against 1.58, and catalog purchases reach 3.03 against 1.78

Table 6. From findings to data improvements

Finding	Data improvement tested
Extreme values in income, web visits, catalog purchases and other continuous features	1. Cap continuous features at the 1st and 99th percentile (variant: cap only spend and income features)
Responders are 14.6% of customers and 46% of them are missed at the 0.5 cutoff	2. Oversample responders in the training data (variant: combined with capping)
Missed responders have no campaign history; false alarms are engaged, recent, high-spending customers	3. Add features aimed at these groups: a no-previous-acceptance flag and two interaction features
Skew is mild and the features are standardized	Log transform considered and not applicable

4. Three Data Improvements

Each improvement changes the data, not the model. Caps and feature definitions are learned on training data only and then applied unchanged to the validation and test data. Each was scored with the same 15 cross-validation folds (5 folds repeated 3 times) on the training data, with fitting done inside each fold, and compared with the baseline using paired differences.

1. Cap extreme values. Values below the 1st or above the 99th percentile of each of the 18 continuous features are set to those limits. Binary and one-hot columns are not touched. This changed 278 of 23,148 training values (1.2%).

2. Oversample responders. Responders in the training data are randomly repeated until they are as common as non-responders. The validation and test data keep their real response rate.

3. Add targeted features. A flag for customers who never accepted a previous campaign, and two interactions (previous acceptances x recency, and total spend x recency), added on top of capping.

Table 7. Cross-validated PR-AUC for the first two improvements and their variants

Data treatment	PR-AUC (mean)	Change vs baseline	Std. error of change	Folds improved (of 15)
Baseline (no data change)	0.7193	0.0000	n/a	n/a
Cap continuous features	0.7237	+0.0044	0.0030	10
Cap continuous + oversample	0.7212	+0.0019	0.0048	8
Cap spend and income features	0.7192	-0.0001	0.0015	6
Oversample responders	0.7173	-0.0020	0.0029	6

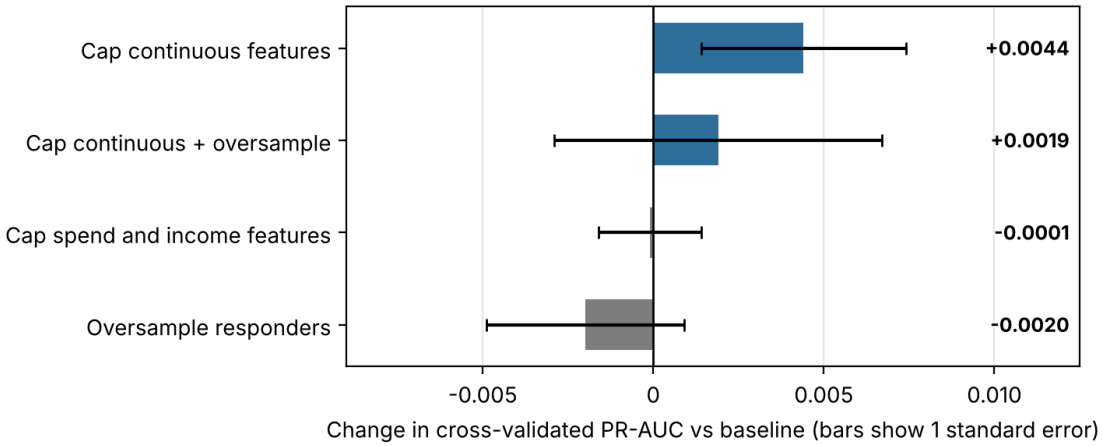


Figure 1. Change in cross-validated PR-AUC versus the baseline for each treatment. Error bars are one standard error of the paired fold differences.

Table 8. Targeted features added on top of capping (third improvement)

Version	PR-AUC (mean)	Change vs capping alone	Std. error	Folds better than capping (of 15)
Capping alone (selected in Table 7)	0.7237	0.0000	n/a	n/a
Cap + two interactions	0.7228	-0.0010	0.0023	7
Cap + flag + interactions	0.7212	-0.0026	0.0038	9

Version	PR-AUC (mean)	Change vs capping alone	Std. error	Folds better than capping (of 15)
Cap + no-acceptance flag	0.7188	-0.0049	0.0032	5

Results.

- **Capping continuous features was the only improvement over the baseline:** +0.0044 PR-AUC, better on 10 of 15 folds and 1.5 standard errors above zero. Folds share training data, so the standard error is optimistic and we treat the gain as small. Capping only the spend and income features made no difference (-0.0001), which suggests that any gain comes from the other continuous features, such as web visits, catalog and deal purchases.
- **Oversampling lowered PR-AUC (-0.0020).** This is expected: PR-AUC measures how well customers are ranked, and repeating responders changes the class weights more than the ranking.
- **The targeted features did not beat capping.** The best version was 0.0010 below it, within one standard error. The no-acceptance flag carries no information a linear model does not already have: previous acceptances are a count, and the response rate rises with it (7.2% at 0, 33.7% at 1, 53.2% at 2 and above 80% at 3 or more).

5. Refitting the Week 9 Model on the Updated Training Data

The Week 9 winner, a logistic regression with C = 3 and the liblinear solver, was refit with the same settings on the updated training dataset: the 1,286 training customers with capping applied. The capping limits were learned on these customers and applied unchanged to the validation and test datasets, and the three updated datasets were saved. No model setting was changed: 27 predictors, random seed 42 for the data split and the cross-validation, and the same 15 folds as the earlier comparisons.

Table 9. Effect of capping on the training data (largest changes)

Feature	Maximum before	Maximum after	Values changed
Income	5.119	1.988	26
NumWebVisitsMonth	6.239	1.576	9
NumDealsPurchases	3.644	2.634	11
NumCatalogPurchases	3.031	1.778	12
Total_Purchases	2.686	2.013	15
Income_per_Person	3.825	3.216	26

6. Week 10 Model vs Week 9 Model on the Updated Validation Dataset

The saved Week 9 model was fit on all 1,608 original customers, which include the 322 customers now in the validation dataset, so it cannot be scored on them fairly. The Week 9 model is therefore compared as its settings refit on the 1,286-customer training dataset with no data changes. Both models are scored on the same 322 validation customers.

Table 10. Validation dataset performance

Metric	Week 9 model (original data)	Week 10 model (capped data)	Change
PR-AUC	0.7300	0.7268	-0.0031
ROC-AUC	0.9079	0.9091	+0.0012
Precision (0.5 cutoff)	0.7500	0.7353	-0.0147
Recall (0.5 cutoff)	0.5106	0.5319	+0.0213
F1 (0.5 cutoff)	0.6076	0.6173	+0.0097
Log loss (lower is better)	0.2373	0.2370	-0.0003
Top-40% capture	0.9149	0.9149	0.0000

Which model performed better, and why. On the validation dataset the two models are tied. The Week 10 model is slightly better on ROC-AUC, recall, F1 and log loss, and the Week 9 model is slightly better on PR-AUC and precision. A paired bootstrap of the PR-AUC change gives an interval of -0.021 to +0.011, which includes zero, and the validation dataset has only 47 responders, so differences this small are within noise. They are small because capping changes only 1.2% of the values, and a regularized logistic regression on standardized features is already fairly stable.

Final model: the Week 10 model. We selected the logistic regression trained on the capped data for three reasons. First, the 15-fold cross-validation on the training data is a larger and more reliable comparison than a 47-responder validation set, and it favors capping (+0.0044 PR-AUC, 10 of 15 folds). Second, it is no worse on four of the seven validation metrics and equal on one. Third, it limits the influence of a few extreme customers, such as the 26 extreme income values, on the model coefficients, which makes it less sensitive to unusual customers once it is deployed. The cost is one extra preprocessing step; the capping limits are saved with the model. The test set did not influence this choice.

7. Final Model Performance on the Test Dataset

Table 11. Final model on the training, validation and test datasets

Metric	Training	Validation	Test
PR-AUC	0.7691	0.7268	0.6780

Metric	Training	Validation	Test
ROC-AUC	0.9464	0.9091	0.8923
Precision (0.5 cutoff)	0.7895	0.7353	0.7179
Recall (0.5 cutoff)	0.5615	0.5319	0.4444
F1 (0.5 cutoff)	0.6562	0.6173	0.5490
Log loss (lower is better)	0.2013	0.2370	0.2748
Top-40% capture	0.9733	0.9149	0.8571

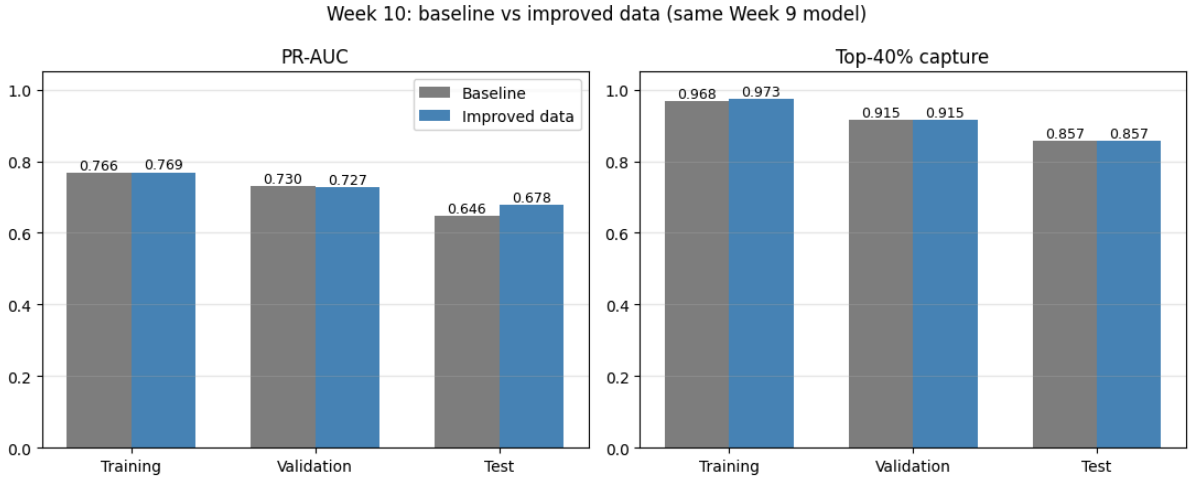


Figure 2. PR-AUC and top-40% capture on the three datasets for the Week 9 model on the original data and the final model on the capped data.

Test compared with validation and training. Every metric gets worse from training to validation to test. PR-AUC falls from 0.769 to 0.727 to 0.678, log loss rises from 0.201 to 0.237 to 0.275, and the share of responders found in the top 40% of customers falls from 97.3% to 91.5% to 85.7%. The drop from training to validation (0.042 PR-AUC) is small. The further drop from validation to test (0.049) is similar in size and is mostly within what chance allows: the test set has 63 responders, and at the 0.5 cutoff the model finds 28 of them (recall 0.444) against 25 of 47 on validation (0.532). The bootstrap interval for the improvement over the baseline on test (+0.001 to +0.082) is wide for the same reason.

For reference, the saved Week 9 model scored a test PR-AUC of 0.665, and the Week 9 settings refit on the smaller training dataset scored 0.646. The final model's 0.678 is the highest of the three, but we do not read the test gain as proof of improvement, because validation shows no gain. On test, precision, recall, F1 and top-40% capture are identical for the Week 9 and Week 10 models, because both find the same 28 responders; only the ranking metrics differ (PR-AUC 0.646 to 0.678, ROC-AUC 0.884 to 0.892, log loss 0.285 to 0.275).

8. Insights from the Training, Validation and Test Errors

1. **The model generalizes reasonably.** The gap between training and validation is modest (PR-AUC 0.769 against 0.727), consistent with the Week 9 finding that logistic regression overfits least. Added flexibility is not the way to improve it.
2. **Expect production performance close to the test result.** Planning should assume a PR-AUC of about 0.68 and about 86% of responders found in the top 40% of customers, not the higher training and validation figures. With 63 test responders these estimates are uncertain, and a monitoring plan should track them on new campaigns.
3. **Precision is stable, recall is the weak side.** Precision stays between 0.72 and 0.79 across the three datasets, so customers the model flags at the 0.5 cutoff usually do respond. Recall falls to 0.44 on test, so many responders are never flagged. A fixed top-share rule, such as contacting the top 40% of customers by predicted probability, finds far more responders than the 0.5 cutoff.
4. **The probabilities can be trusted.** Log loss rises smoothly from training to test and the calibration check is close, so the predicted probabilities can be used directly in the contact-cost decision from the Week 9 business case.
5. **The ceiling is missing information, not data quality.** Cleaning the data gave a small gain at most, and features built from existing columns gave none. About 40% of responders have no previous campaign acceptance and look like ordinary non-responders. Further gains most likely need new information about customers, such as channel or product preferences or the results of earlier offers, together with a larger number of responders to learn from.

9. Conclusion

We kept the Week 9 model fixed and tested three data improvements chosen from the error analysis. Capping extreme values on the continuous features was the only one that helped, by a small amount in cross-validation, and it is part of the final model. On the validation dataset the capped-data model and the original-data model are tied, so the selection rests on the cross-validation result and on robustness to extreme values. The final model reaches a test PR-AUC of 0.678 and finds 85.7% of responders in the top 40% of customers. The remaining errors come mostly from customers with no campaign history, so the next improvement is more information about customers, not further cleaning of the data we have.